MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5656

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Double tetrode, RF, miniature

Outline	---	6-7 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	g2 (note 1)	1g1	2g1	h	h	k (note 2)	2a	1a	k (note 2)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	(A) Ehk	(A) Ib/a	(A) Ik	Ic1/g	Pp/a	Pg2	Pi	tk	(A) Alt
Unit:	V	Vdc	Vdc	Vdc	v	mAdc	mAdc	mAdc	W	W	W	sec(min)	ft
Maximum:													
Class C Teleg intermittent duty:	6.9	250	-50	165	100	25	65	4	3.5	1.8	11	30	See note 3
Class C Teleg continuous duty:	6.9	220	-50	165	100	17.5	65	4	2.5	1.5	7	30	See note 3
Class A Audio and RF duty:	6.9	250	---	165	100	20	65	---	3.0	1.5	---	30	See note 3
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	-2.0	120	---	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

(A) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See notes 4 and 5	0.65	II	Ic1	0	-1.0	μAdc
1301	Heater current		0.65	II	If	360	440	mA
1256	Electrode current (screen)	See note 4	0.65	II	Ic2	0	4.0	mAdc
1256	Electrode current (1) (anode)	See note 4	0.65	II	Ib	9	21	mAdc
1256	Electrode current (2) (anode)	Ec1 = -12 Vdc (see note 4)	0.65	II	Ib	---	200	μAdc
1306	Transconductance	See note 4	0.65	II	Sm	4,500	7,500	μmhos
1236	Power oscillation (1)	Eb = 220 Vdc; Ec2/Ib = 45 mAdc; Ic1/g = 3 mAdc; Rg1/g = 5,000 ohms; F = 225 MHz; Ef = 6.3 Vac; push-pull amplifier	0.65	II	Po	4.0	---	W
1236	Power oscillation (2)	Power oscillation (1); Ef = 5.7 Vac; t = 60 sec	0.65	II	Po	3.0	---	W
1231	Peak emission	eb = ec1 = ec2 = 75 v; (see notes 4 and 6)	0.65	II	is	250	---	ma
(A) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 10,000 ohms; (see note 4)	---	---	Ep	---	1,000	mVac
(A) 1336	Heater-cathode leakage		---	---	Ihk	---	20	μAdc
1236	Power oscillation (3)	Eb = 220 Vdc; Ecc2 = 220 Vdc; Rg2 = 8,200 ohms; Rg1 = 5,000 ohms; Ic1 = 2.5 mAdc; Ib = 45 mAdc; Ef = 5.6 Vac; F = 390 MHz; t = 300 sec; push-pull amplifier	---	---	Po	1.5	---	W
1331	Direct-interelectrode capacitance	No shield (see note 7)	---	---	Cgp Cin Cout	--- 2.7 1.0	0.06 4.5 2.0	pF pF pF
(A) 1121	Base strain	See note 8	---	---	---	---	---	---
(A) 2126	Glass strain		4.0	I	---	---	---	---
(A) 1105	Permanence of marking		---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
Ⓐ ---	Life-test provisions	Group B; 225 MHz push-pull Class C self oscillator; Ebb = 220 Vdc; Ec2/Ik = 60 mAdc; Ef = 6.3 Vac; Ic1/g = 3.0 mAdc; Rg1/g = 3,000 ohms; Rk = 100 ohms; Rb = 750 ohms; TA = room; intermittent operation, Ef only 1-hour "on", (Ef, Ebb, and Ec2 all "on") oscillation 1-hour "on", t = total oscillation and Ef only time.	---	---	---	---	---	---
---	Life-test end point (500 hours)	Power oscillation (1)	---	---	Po	3.0	---	W

NOTES:

1. The screen grids for the two sections are connected internally and by-passed to cathode by an internal capacitor of approximately 25 pF.
2. Cathodes are connected internally.
3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
4. Read each unit separately. Control grid of unit not under test shall be connected to -45 volts.
5. This test to be performed at the conclusion of the holding period.
6. Voltage shall be applied in pulses such that the tube will not be damaged.
7. Duplicate test on each unit separately.
8. Sampling inspection shall be in accordance with MIL-E-1 paragraph "Base-strain test, miniature tube, sampling (method 1121)."

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-8)

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:21Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:21Z
Check the source to verify that this is the current version before use.